

GSSSP : Generator of Synthetic Scans of Spectroscopic Plates for Object Detection

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Abstract. GSSSP (Generator of Synthetic Scans of Spectroscopic Plates) is a tool for generating labeled scans of physical spectroscopic observations. The area of spectroscopic analysis is old in the Astronomic Domain. Legacy observational methods recorded stellar spectra on photographic plates. These plates have high historical and scientific value. However, a limited number of them have been digitized, resulting in an insufficient amount of data to train modern computer vision models for observation detection.

In this work, we present a software tool that replicates the appearance of scanned spectroscopic plates by generating synthetic images containing spectra of artificial stellar objects. The generated images are accompanied by label files that enable the training of observation detection models.

We trained a detection model and compared its performance when using only real data, only generated data, and a combination of both. The results show that GSSSP enables the model to learn the general visual patterns of this type of plate, providing a strong generalization basis for subsequent fine-tuning. The resulting model surpasses the current state of the art for this problem.

Keywords: Generated data, Object detection, Spectroscopic plates

1 Spectroscopic Data

Descripcion de las placas, relevancia, contexto de que hay pocas.

1.1 Available Data

For this study, we use a dataset composed of 361 spectroscopic plates from **Cerro Tololo** that were scanned and labeled for observation detection. The dataset is publicly available ⁵. Due to the limited availability of digitized spectroscopic plates, the dataset size is relatively small.

2 State of Art

Mencion a trabajos previos donde se construyeron modelos de deteccion para observaciones.

3 GSSSP

En que consiste GSSSP, sus modulos. Caracteristicas principales, personalizacion de observaciones e imagenes de espectros.

4 Experiment

Detalles del entrenamiento, datos usados en cada uno y metricas de evaluacion.

4.1 Results

Resultados de cada entrenamiento.

5 Conclusion

Conclusions.

6 Sample Section

6.1 A Subsection Sample

Please note that the first paragraph of a section or subsection is not indented. The first paragraph that follows a table, figure, equation etc. does not need an indent, either.

Subsequent paragraphs, however, are indented.

⁵ Observation Detection in Spectroscopic Plates Dataset

Sample Heading (Third Level) Only two levels of headings should be numbered. Lower level headings remain unnumbered; they are formatted as run-in headings.

Sample Heading (Fourth Level) The contribution should contain no more than four levels of headings. Table 1 gives a summary of all heading levels.

Table 1. Table captions should be placed above the tables.

Heading level	Example	Font size and style
Title (centered)	Lecture Notes	14 point, bold
1st-level heading	1 Introduction	12 point, bold
2nd-level heading	2.1 Printing Area	10 point, bold
3rd-level heading	Run-in Heading in Bold. Text follows	10 point, bold
4th-level heading	<i>Lowest Level Heading.</i> Text follows	10 point, italic

Displayed equations are centered and set on a separate line.

$$x + y = z \tag{1}$$

Please try to avoid rasterized images for line-art diagrams and schemas. Whenever possible, use vector graphics instead (see Fig. 1).

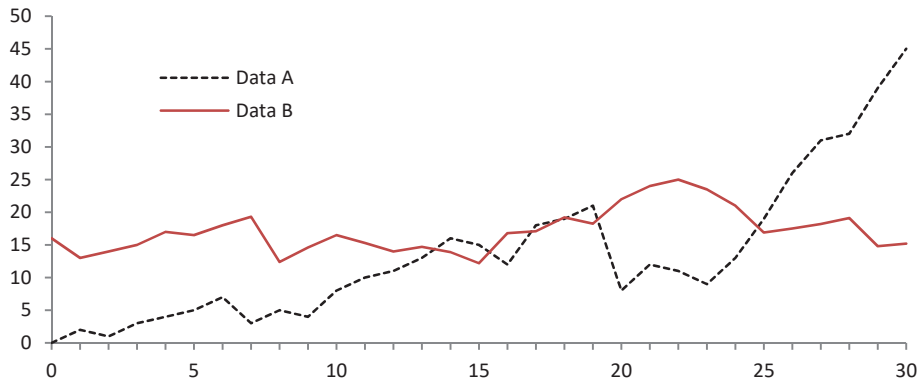


Fig. 1. A figure caption is always placed below the illustration. Please note that short captions are centered, while long ones are justified by the macro package automatically.

Theorem 1. *This is a sample theorem. The run-in heading is set in bold, while the following text appears in italics. Definitions, lemmas, propositions, and corollaries are styled the same way.*

Proof. Proofs, examples, and remarks have the initial word in italics, while the following text appears in normal font.

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable. The following bibliography provides a sample reference list with entries for journal articles [1], an LNCS chapter [2], a book [3], proceedings without editors [4], and a homepage [5]. Multiple citations are grouped [1,2,3], [1,3,4,5].

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